Ashutosh Tiwari

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Work Experience

Senior Machine Learning Engineer (ML Platform & Frameworks) Jul. 2024 – Present
ADOBE FIREFLY *GENERATIVE AI, COMPUTER VISION, LARGE LANGUAGE MODELS*

- Leading part of ML Platform, focused on offline inference for videos and images, and training frameworks.
- Designed and led engineering efforts for distributed inference frameworks: online (vLLM + Ray for LLMs like Falcon-40B, Llama 2 70B) for experimentation and offline inference (Ray Data + PyTorch Lightning, achieving 10-50x faster image loading, 3-8K images/sec on 32 GPUs).
- Leading development of PyTorch based training framework for large-scale foundation models, adopted across the organization for training generative models on billions of images and videos. Core contributions include a plugin-based strategy pattern over FSDP/FSDP2, and integrations for Flash Attention 2/3, context parallelism, and distributed attention for DiT text-to-image and text-to-video training.
- Leading enrichment service with fire and forget pipelines with dataset sizes of billions of images and videos, which is an abstraction and orchestration layer on inference framework and other ML platform offerings.
- Wrote the first data quality framework used to ensure the quality of feature generation using Cerberus. This is used across enrichment pipelines to ensure the correctness of output.

Senior Software Engineer (Data Platform) Sep. 2023 – Jul. 2024
EVOLUTIONIQ *DATA PLATFORM, NATURAL LANGUAGE PROCESSING, REGRESSION*

- Part of Scoring and Quality sub-team, working at the intersection of generative AI, fin-tech, and health sector.
- Led the development of data pipelines that ingest, and process data to be ingested, and used for training by our machine learning models.
- Led the development of pipelines that train our models to predict a claimant's expected time to return to work, ICD extractions from diagnosis notes, alternate VOC recommendations, etc.
- Worked on a framework to evaluate and measure model fairness and bias metrics of our models towards different demographic groups.

Software Dev Engineer II (ML Platform) Jan. 2019 – Jul. 2021
SWIGGY *DATA SCIENCE PLATFORM, TIME SERIES FORECASTING*

- Built and maintained Feature Store and pipeline feeding on-demand features to deployed ML models at production scale (4Bn rows, 10K QPS). Pipeline supported multichannel ingestion, i.e., Spark, Flink and user files etc.
- Core contributor to the online inference platform serving 18M+ monthly transacting users across food delivery and quick commerce. Models were optimized for latency and throughput through autoscaling policies that drove cluster scale-outs based on real-time load metrics.
- Engineered a dynamic request routing layer where incoming API calls were evaluated against geo-tags, traffic profiles, and service metadata, and then dispatched to the appropriate model clusters partitioned by region and load domain.
- Implemented the routing/dispatch substrate on an Akka-style actor framework to achieve resilient, asynchronous model invocation, fault isolation across clusters, and backpressure-aware scheduling for sustained low P99 latency under high QPS.
- Founding member of Forecasting and Correlation Platform which was considered by many teams to forecast concerned time series. These forecasts power critical scaling decisions across organizations in real time.
- Led DAQ, a tool used to scrape APIs at scale. Used to collect data for analysis/model training at a scale of 15M rows daily.

Software Development Engineer (Search Relevance) Sep. 2017 – Jan. 2019
FLIPKART (a Walmart company) *SEARCH RELEVANCE, QUERY INTENT, NLP*

- Improved and built search intent models (CRF/Neural Network based), identifying error classes, designing solutions, and shipping fixes. These models power user search and discovery for millions every day.
- Implemented a FastText based query store classifier, which predicts the category of a tail query.
- Implemented the first workflow to automate training and auto-deployment of various search models in Flipkart. First was written using Luigi and later migrated to Airflow.
- Wrote a generic framework using Airflow which at runtime creates generic dags for different ML models and orchestrates their training to deployment flow, including data and model validations.
- Implemented large scale (4Bn+ datapoints) pipelines using Cascading/HDFS to extract data from user events and then transform it to be used for training these models.

Software Development Engineer Sep. 2016 – Sep. 2017
GROUPON *BACKEND ENGINEERING*

- Worked on a component called Cyclops, an interface between Customer representatives and internal services.
- This service is live in all countries in which Groupon operates.

Software Engineer Sep. 2015 – Aug. 2016
NETSPEED SYSTEMS (Acquired by Intel) *GRAPH ALGORITHMS, NETWORK ON CHIP*

- Led engineering efforts on modules like Polarity based Arbitration, Multi-Cast Filtering, Structural Latency Breakdown, etc.

Publications

Accepted at *NetSci 2023* (Poster Presentation) and *IC2S2 2023* (Parallel Talk) as **first author**

[1] **Ashutosh Tiwari**, Prof. Sadamori Kojaku, Prof. Yong-Yeol Ahn, "Biased Contrastive Learning debiases Graph Neural Networks," *International Conference on Network Science (NetSci)*, 2023. Poster

In this work, we propose a non-parametric contrastive learning framework to learn debiased graph embeddings with respect to sensitive node attributes and structural homophily. Through empirical evaluations on different datasets, we demonstrate that our method offers a better approach to debiasing compared to existing approaches and thus results in more organic recommendations across different GNN architectures.

Education

Indiana University, Bloomington — Bloomington, IN
Master of Science in Computational Data Science 3.87/4.0 — *Aug. 2021 – May 2023*

National Institute of Technology — Patna, India
Bachelor of Science in Computer Science 8.32/10.0 — *Aug. 2011 – May 2015*

Research Experience / Independent Study

- Implementing novel training frameworks that result in unbiased (or fair) recommendation systems from "biased" datasets at IUNI with Prof. YY Ahn and Prof. S Kojaku as part of my Independent Study. Summer 2022 - Summer 2023
- Working as a paid RA for, **"User Intent as a Network"** with Prof. YY Ahn, P Kantak and FB Yara. Project is funded by Kelly Business School. Goal is to be able to quantify and thus act upon that "intent" network. Summer 2022 - Fall 2022
- Was part of NLP Lab@IUB with Prof. D Cavar. Contributed to design of TieML and Events' Timeline modelling using different fine-tuned Large Language Models (LLMs). Fall 2021

Teaching Experience

- Teaching Assistant for Network Science (INFO-I 606) with Prof. YY Ahn — Spring 2023
- Teaching Assistant for Machine Learning (CSCI-B 555) with Prof. R Khardon — Fall 2022
- Teaching Assistant for Network Science (INFO-I 606) with Prof. YY Ahn — Spring 2022

Selected Projects

BiasNet | *Deep Reinforcement Learning, PyTorch, Actor Critic Algorithm, On-policy Model Free* — Code/Report
- Learning to fight in Street Fighter II with induced relational bias from differential game scenes.

DeepFoodie | *Python, Tensorflow, Self Supervised Deep Clustering, Deep Learning, Transfer Learning* — Code/Report
- Clustering dishes on basis of their ingredient embeddings. These ingredient embeddings are generated by a NN.

BlindNet | *Python, Pytorch, Deep Learning, Transfer Learning* — Code/Report
- Image to vector generation on Coco dataset.

Continuous Dominant Set Repair | *C++, Graph Algorithms, Guha and Khuller's Algo., Greedy* — Code
- Repairs a broken link in Continuous Dominant Set in $O(\Delta^2)$, where Δ being the avg cardinality of connected graph.

Humana Mays Healthcare Analytics Case Competition | *Boosted Trees, Feature Engineering* — Code
- 11th rank on leaderboard. Hosted by TAMU and Humana Mays, 2021.

AnalyticsVidhya Solutions | *Python, Tensorflow, Torch, Time Series, Feature Engineering, Catboost* — Code

Investigating Bias Manifolds | *Bias Manifolds, Bias Progression, Measuring Bias, Python, Word2vec* — Code/Report

Technical Skills

Search Relevance, Recommendations, Graph Neural Networks, Fairness Aware Modeling, Computational ML, NLP, Computer Vision, DL, Spark, Python, Scala, C++, Contrastive Learning, Large Language Models
Frameworks/Libraries/Tools: Ray, Pytorch Geometric, Pytorch Lightning, Pytorch, Tensorflow, Sklearn, Numpy, Pandas, Matplotlib, HDFS, Spark, Kafka, Flink, AWS Sagemaker, DynamoDB, Faiss, vLLM, Django